

Multiple Choice (10 marks)

1. What is 836 rounded to the nearest 10?
 - (a) 800
 - (b) 830
 - (c) 840
 - (d) 870
2. A point in space (x, y, z) is randomly selected so that $-1 \leq x \leq 1, -1 \leq y \leq 1, -1 \leq z \leq 1$. What is the probability that $x^2 + y^2 + z^2 \leq 1$?
 - (a) $\frac{\pi}{24}$
 - (b) $\frac{\pi}{12}$
 - (c) $\frac{\pi}{6}$
 - (d) $\frac{\pi}{3}$
3. Express $0.1\overline{7}$ as a common fraction.
 - (a) $\frac{8}{45}$
 - (b) $\frac{8}{5}$
 - (c) $\frac{8}{7}$
 - (d) $\frac{8}{33}$
4. Express as a common fraction the reciprocal of $0.\overline{7}$.
 - (a) $\frac{7}{2}$
 - (b) $\frac{7}{9}$
 - (c) $\frac{9}{2}$
 - (d) $\frac{9}{7}$
5. What is the smallest two-digit integer n such that switching its digits and then adding 3 results in $2n$?
 - (a) 6
 - (b) 12
 - (c) 96
 - (d) 102

True / False (10 marks)

Answer True or False

6. True or False: -4 plus -3 equals -1 .
7. True or False: The 5th term in the series, starting with 2 and multiplying by -2 each time, is -16 .
8. True or False: The median of the data set 13, 35, 26, 8, 24, 10, 22, 10, 32 is 20.
9. True or False: A simple random sample is determined by how representative the sample is of the entire population.
10. True or False: If 7 coins represent 5% of Kate's collection, then her total coin collection is 120 coins.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. In how many ways can we form an international commission if we must choose one European country from among 6 European countries, one Asian country from among 4, one North American country from among 3, and one African country from among 7?
12. Discuss how increasing the sample size from 200 to 2,000 in a study of geese returning to the same breeding site affects the distribution of the sample proportion. Provide a detailed explanation.

End of Paper